AP CALCULUS AB

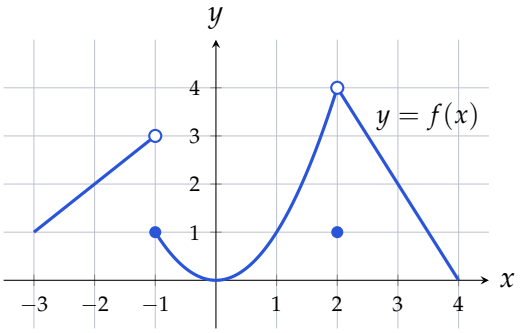
Unit 1 • Lesson 1.10 practice • Answer key

Lesson 1.10 • Exploring Types of Discontinuities

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



The graph of f: a line segment for $-3 \le x < -1$, $y = x^2$ for $-1 \le x < 2$, a line segment for $2 < x \le 4$; $f(-1) = 1$, $f(2) = 1$.

Question 1

NO CALCULATOR

What type of discontinuity does $f(x) = \dfrac{x^2 - 1}{x - 1}$ have at $x = 1$?

(A) Jump
(B) Infinite
(C) Removable
(D) None

Solution

Answer: (C). $\lim_{x \to 1} f(x) = 2$ exists, but $f(1)$ is undefined.

Question 2

NO CALCULATOR

$\lim_{x \to 0^-} g(x) = 1$ and $\lim_{x \to 0^+} g(x) = 4$. What type of discontinuity does g have at $x = 0$?

(A) Jump
(B) Removable
(C) Infinite
(D) None

Solution

Answer: (A). The one-sided limits exist and are different.

Question 3

NO CALCULATOR

What type of discontinuity does $h(x) = \dfrac{1}{x-2}$ have at $x = 2$?

(A) Removable
(B) Infinite
(C) Jump
(D) None

Solution
Answer: (B). The one-sided limits are $-\infty$ and $+\infty$.

Question 4

NO CALCULATOR

For the function f graphed above, which describes its discontinuities at $x = -1$ and $x = 2$?

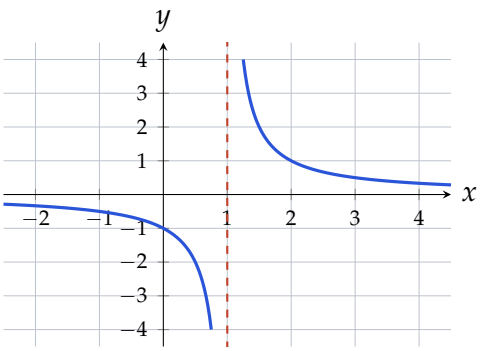
(A) Both removable
(B) Both jumps
(C) Removable at -1, jump at 2
(D) Jump at -1, removable at 2

Solution
Answer: (D). At -1 the one-sided limits are 3 and 1; at 2 the limit is 4 but $f(2) = 1$.

Question 5 (graph)

NO CALCULATOR

The graph of h is shown. What type of discontinuity does h have at $x = 1$?



(A) Removable
(B) Jump
(C) Infinite
(D) None; h is continuous at $x = 1$.

Solution
Answer: (C). $\lim_{x\to 1^-} h(x) = -\infty$ and $\lim_{x\to 1^+} h(x) = +\infty$: an infinite discontinuity.

Question 6 (Free response)

NO CALCULATOR

Let $g(x) = \begin{cases} x^2, & x < 0 \\ x + 1, & 0 \le x \le 2 \\ \dfrac{1}{x-3}, & x > 2. \end{cases}$

(a) Find the one-sided limits of g at $x = 0$ and classify the discontinuity.
(b) Find the one-sided limits of g at $x = 2$ and classify the discontinuity.
(c) Classify the discontinuity at $x = 3$. Justify with limits.

Solution
(a) $\lim_{x\to 0^-} g(x) = 0$ and $\lim_{x\to 0^+} g(x) = 1$: jump discontinuity.
(b) $\lim_{x\to 2^-} g(x) = 3$ and $\lim_{x\to 2^+} g(x) = \frac{1}{2-3} = -1$: jump discontinuity.
(c) $\lim_{x\to 3^-} \frac{1}{x-3} = -\infty$ and $\lim_{x\to 3^+} \frac{1}{x-3} = +\infty$: infinite discontinuity (vertical asymptote $x = 3$).